



# **Lab Sheet 1**

## **Database Management System Lab (BTCS203P)**

**B. TECH AI&ML, CYBER SECURITY, Data Science 3<sup>RD</sup> Semester (Session 2026-2025)**

### **Objective:**

To understand and practice Data Definition Language (DDL) commands for creating and modifying the structure of a table in a relational database using SQL (MySQL).

### **Prerequisites:**

1. Basic knowledge of SQL.
2. MySQL installed and set up.

### **Task : Open SQL client and perform the following tasks**

#### **Task 1: Create a Table**

a. Create a database named Company.

= CREATE DATABASE Company;

b. Create a table named Employees with the following structure:

| Column Name             | Data Type     | Constraints    |
|-------------------------|---------------|----------------|
| EmpID<br>AUTO_INCREMENT | INT           | PRIMARY<br>KEY |
| FirstName               | VARCHAR(50)   | NOT NULL       |
| LastName                | VARCHAR(50)   |                |
| Department              | VARCHAR(50)   |                |
| Salary                  | DECIMAL(10,2) |                |
| Email                   | VARCHAR(100)  | UNIQUE         |

=USE Company;

```
CREATE TABLE Employees (  
    EmployeeID INT AUTO_INCREMENT PRIMARY KEY,  
    FirstName VARCHAR(50) NOT NULL,  
    LastName VARCHAR(50),  
    Department VARCHAR(50),  
    Salary DECIMAL(10,2),  
    Email VARCHAR(100) UNIQUE  
);
```

#### **Task 2: DDL Queries on the Employees Table**

a) Describe the structure of the table.

=DESCRIBE Employees;

**b)** Rename the table to Staff.  
 =RENAME TABLE Employees TO Staff;

**c)** Add a new column PhoneNumber (VARCHAR(15)).  
 =ALTER TABLE Staff  
 ADD PhoneNumber VARCHAR(15);

**d)** Modify the data type of Salary to DECIMAL(12,2).  
 =ALTER TABLE Staff  
 MODIFY LastName VARCHAR(50);

**e)** Add a DEFAULT value of 'General' to the Department column.  
 =ALTER TABLE Staff  
 ALTER COLUMN Department SET DEFAULT 'General';

**f)** Drop the Phone Number column.  
 =ALTER TABLE Staff  
 DROP COLUMN PhoneNumber;

**g)** Add a CHECK constraint to ensure Salary is greater than 0.  
 =ALTER TABLE Staff  
 ADD CONSTRAINT chk\_Salary CHECK (Salary > 0);

**h)** Rename the column First Name to Given Name.  
 =ALTER TABLE Staff  
 RENAME COLUMN FirstName TO GivenName;

**i)** Drop the chk\_Salary constraint:  
 =ALTER TABLE Staff  
 DROP CHECK chk\_Salary;

**j)** Create an index on the Email column.  
 =CREATE INDEX idx\_email  
 ON Staff(Email);

**k)** Drop the index on the Email column.  
 =DROP INDEX idx\_email ON Staff;

**l)** Truncate the table to remove all rows without affecting the structure.  
 =TRUNCATE TABLE Staff;

**m)** Drop the Staff table completely  
 =DROP TABLE Staff;

### 3. Conceptual Questions:

1. What is the difference between TRUNCATE and DELETE?  
 =**TRUNCATE** removes **all records** from a table at once, while **DELETE** can remove specific records using a WHERE condition.

- TRUNCATE cannot normally use WHERE.
- DELETE can use WHERE.
- TRUNCATE is generally faster for removing all rows.
- Both keep the table structure.

#### Example:

TRUNCATE TABLE Employees;

DELETE FROM Employees WHERE EmployeeID = 5;

2. Explain the use of the CHECK constraint.

=The **CHECK constraint** is used to ensure that the values entered into a column satisfy a specific condition. For example:

Salary DECIMAL(10,2), CHECK (Salary > 0)

This ensures that the salary cannot be **zero or negative**. It helps maintain **valid and accurate data** in the table.

3. Why is it important to define a PRIMARY KEY for a table?

=A **PRIMARY KEY** uniquely identifies each record in a table.

For example:

Employee ID INT PRIMARY KE

Importance:

- It prevents duplicate values.
- It does not allow NULL values.
- It uniquely identifies each row.
- It helps maintain **data integrity**.
- It makes it easier to reference a particular record.

4. How does ALTER TABLE help in modifying an existing table?

=ALTER TABLE is used to **change the structure of an existing table** without creating a new table.

- Add a new column.
- Delete a column.
- Modify a column's data type.
- Rename a column.
- Add or remove constraints.

EXAMPLE:

ALTER TABLE Employees ADD PhoneNumber VARCHAR(15);

This adds a new Phone Number column to the existing table.

5. What is the purpose of creating an index in a table?

=An **index** is created to make searching and retrieving data from a table **faster**.

EXAMPLE:

CREATE INDEX idx\_email ON Employees(Email);

This creates an index on the Email column, which can improve the speed of queries that search using that column.

#### 4. Exercises:

1. Create a table named Projects with the following structure:

a) Perform at least 5 DDL operations on the Projects table.

| Column Name | Data Type     | Constraints                |
|-------------|---------------|----------------------------|
| ProjectID   | INT           | PRIMARY KEY AUTO_INCREMENT |
| ProjectName | VARCHAR(100)  | NOT NULL                   |
| StartDate   | DATE          |                            |
| EndDate     | DATE          |                            |
| Budget      | DECIMAL(15,2) | CHECK (Budget > 0)         |

```
=CREATE TABLE Projects (
  ProjectID INT AUTO_INCREMENT PRIMARY KEY,
  ProjectName VARCHAR(100) NOT NULL,
  StartDate DATE,
  EndDate DATE,
  Budget DECIMAL(15,2),
  CONSTRAINT chk_Budget CHECK (Budget > 0)
);
DESCRIBE projects;
```

```
ALTER TABLE Projects
ADD ManagerName VARCHAR(100);
ALTER TABLE Projects
MODIFY ProjectName VARCHAR(150) NOT NULL;
ALTER TABLE Projects
RENAME COLUMN ManagerName TO ProjectManager;
ALTER TABLE Projects
DROP COLUMN ProjectManager;
CREATE INDEX idx_project_name
ON Projects(ProjectName);
DROP INDEX idx_project_name ON Projects;
```

**b)** Write a query to retrieve the names of all columns in the Staff table.

```
=SELECT COLUMN_NAME
FROM INFORMATION_SCHEMA.COLUMNS
WHERE TABLE_SCHEMA = DATABASE()
AND TABLE_NAME = 'Staff';
DESCRIBE Staff;
```

**c)** Write a query to modify the Department column to allow NULL values.

```
=Department VARCHAR(50) NOT NULL
ALTER TABLE Staff
MODIFY Department VARCHAR(50) NULL;
```

Limit to 1000 rows

```

1 • DROP DATABASE IF EXISTS company;
2 • CREATE DATABASE Company;
3 • USE Company;
4 • CREATE TABLE Employee(
5     EmpID INT AUTO_INCREMENT PRIMARY KEY,
6     FirstName VARCHAR(50) NOT NULL,
7     LastName VARCHAR(50),
8     Department VARCHAR(50),
9     Salary DECIMAL(10,2),
10    Email VARCHAR(100) UNIQUE );
11 • DESCRIBE Employee;
12 • RENAME TABLE Employee TO Staff;
13 • Alter TABLE Staff

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

| Field      | Type          | Null | Key  | Default              | Extra          |
|------------|---------------|------|------|----------------------|----------------|
| EmpID      | int           | NO   | PRI  | <a href="#">NULL</a> | auto_increment |
| FirstName  | varchar(50)   | NO   |      | <a href="#">NULL</a> |                |
| LastName   | varchar(50)   | YES  |      | <a href="#">NULL</a> |                |
| Department | varchar(50)   | YES  |      | <a href="#">NULL</a> |                |
| Salary     | decimal(10,2) | YES  |      | <a href="#">NULL</a> |                |
| Email      | varchar(100)  | YES  | U... | <a href="#">NULL</a> |                |

Query 1 | SQL File 3\* | SQL File 4\* | SQL File 5\* x

Limit to 1000 rows

```

1 • DROP TABLE IF EXISTS Projects;
2 • CREATE TABLE Projects (
3     ProjectID INT AUTO_INCREMENT PRIMARY KEY,
4     ProjectName VARCHAR(100) NOT NULL,
5     StartDate DATE,
6     EndDate DATE,
7     Budget DECIMAL(15,2),
8     CONSTRAINT chk_Budget CHECK (Budget > 0)
9 );

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

| Field       | Type          | Null | Key | Default              | Extra          |
|-------------|---------------|------|-----|----------------------|----------------|
| ProjectID   | int           | NO   | PRI | <a href="#">NULL</a> | auto_increment |
| ProjectName | varchar(100)  | NO   |     | <a href="#">NULL</a> |                |
| StartDate   | date          | YES  |     | <a href="#">NULL</a> |                |
| EndDate     | date          | YES  |     | <a href="#">NULL</a> |                |
| Budget      | decimal(15,2) | YES  |     | <a href="#">NULL</a> |                |

Limit to 1000 rows

```

20 • CREATE INDEX idx_project_name
21 ON Projects(ProjectName);
22 • DROP INDEX idx_project_name ON Projects;
23 • SELECT COLUMN_NAME
24 FROM INFORMATION_SCHEMA.COLUMNS
25 WHERE TABLE_SCHEMA = DATABASE()
26 AND TABLE_NAME = 'Staff';
27 • DESCRIBE Staff;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

| Field      | Type          | Null | Key  | Default              | Extra          |
|------------|---------------|------|------|----------------------|----------------|
| EmpID      | int           | NO   | PRI  | <a href="#">NULL</a> | auto_increment |
| FirstName  | varchar(50)   | NO   |      | <a href="#">NULL</a> |                |
| LastName   | varchar(50)   | YES  |      | <a href="#">NULL</a> |                |
| Department | varchar(50)   | YES  |      | General              |                |
| Salary     | decimal(12,2) | YES  |      | <a href="#">NULL</a> |                |
| Email      | varchar(100)  | YES  | U... | <a href="#">NULL</a> |                |

```

12 • ALTER TABLE Projects
13 ADD ManagerName VARCHAR(100);
14 • ALTER TABLE Projects
15 MODIFY ProjectName VARCHAR(150) NOT NULL;
16 • ALTER TABLE Projects
17 RENAME COLUMN ManagerName TO ProjectManager;
18 • ALTER TABLE Projects
19 DROP COLUMN ProjectManager;
20 • CREATE INDEX idx_project_name
21 ON Projects(ProjectName);
22 • DROP INDEX idx_project_name ON Projects;
23 • SELECT COLUMN_NAME
24 FROM INFORMATION_SCHEMA.COLUMNS
25 WHERE TABLE_SCHEMA = DATABASE()
26 AND TABLE_NAME = 'Staff';
27 • DESCRIBE Staff;
28 • Department VARCHAR(50) NOT NULL
29 ALTER TABLE Staff
30 MODIFY Department VARCHAR(50) NULL;

```

Output

Action Output

| #  | Time     | Action                                                                               | Message                                                 |
|----|----------|--------------------------------------------------------------------------------------|---------------------------------------------------------|
| 25 | 18:40:23 | ALTER TABLE Projects DROP COLUMN ProjectManager                                      | 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0  |
| 26 | 18:40:23 | CREATE INDEX idx_project_name ON Projects(ProjectName)                               | 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0  |
| 27 | 18:40:23 | DROP INDEX idx_project_name ON Projects                                              | 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0  |
| 28 | 18:41:16 | SELECT COLUMN_NAME FROM INFORMATION_SCHEMA.COLUMNS WHERE TABLE_SCHEMA = DA...        | 6 row(s) returned                                       |
| 29 | 18:41:16 | DESCRIBE Staff                                                                       | 6 row(s) returned                                       |
| 30 | 18:42:15 | Department VARCHAR(50) NOT NULL ALTER TABLE Staff MODIFY Department VARCHAR(50) NULL | Error Code: 1064. You have an error in your SQL syntax; |